Report on Locomotive Economy Devices

The committee on New Locomotive Economy Devices this year reviewed briefly the various devices which are being closely studied by railways in order to institute fuel economies. The proper, intelligent application, use and upkeep of economy devices must be carefully investigated before logical conclusions can be made as to the over-all economy of such devices.

The feed water heaters, pump type, and exhaust feed water heater injectors, have previously been reported upon, and those installed and on order in America to date are as follows, as of May 1, 1926.

	Exhaust Steam Pump Type	Feed Water Heaters Injector Type
1920	7	..
1921	54	..
1922	234	..
1923	1,429	..
1924	2,123	24
1925	2,551	37

There have been no particular developments during the past year in feed water heaters of pump types, either open or closed, other than minor adjustments and corrosion studies.

The Elesco exhaust heater injector has increased in number the past year and has been described in full in the 1925 report. Reports still show an over-all saving, the same as previously given.

Sellers Feed Water Heater Injector

A new exhaust feed water heater injector has completed its experimental stage and is manufactured by William Sellers and Company, Inc., of Philadelphia.

In this system there is a simplicity of design and mechanical operation easily understood by engine men and applicable to the pooling system, the injector principle; a primary set of nozzles is operated by exhaust steam, utilizing the latent heat, and energy to deliver a strong and heated supply of water to a forcing set of nozzles, actuated by steam from the boiler; by those nozzles it is forced through the feed pipe into the boiler. This feed water heater injector reclaims the heat in the exhaust steam from the cylinders; and the heat in the live steam taken from, is also restored to the boiler, resulting in a final delivery temperature of the feed water which may be not far below the temperature of the steam.

The injector is operated by a simple cab stand mechanism. It consists of a lever acting through a link and extension rod, which opens, closes or adjusts the exhaust steam admission valve; a lever handle for opening, closing or adjusting a new form of water-tight, cylindrical lazy cock, a hand wheel for the final overflow. The method of manipulating the Sellers, "V. C." starting valve and cab stand levers, is similar to the Sellers Class K, non-lifting injector, so that it presents no difficulty or uncertainty to an engineer or fireman. The operator by the cab stand is entirely mechanical, consisting of a simple combination of levers, links and universal joints, for which repairs can be made by the ordinary railroad shop mechanic; there are no pneumatic valves or delicate pieces of mechanism. Exhaust steam is obtained from a bridge connecting the cylinders to a regulating valve; and is carried by a flanged pipe with easy bends to the injector, bolted to a bracket on the mudring on the right hand side. The regulating valve and its accessories include: a baffle plate to reduce the pulsations of the exhaust; a balanced piston valve to separate the injector exhaust supply pipe from the cylinders when the throttle is closed; if the injector is feeding while the engine throttle is closed; if the injector is feeding while the engine throttle is closed, this valve also regulates the quantity of exhaust steam to suit the requirements of the injector; an admission valve to substitute live for exhaust steam when the engine is standing or drifting. The operation of these devices is automatic. Within or directly in front of the cab is placed a duplex balanced lever starting valve V-C type, supplying the forcing set of nozzles of the injector, and to the live steam admission section of the regulating valve. In the cab is also placed the Sellers tell-tale or injector indicator, which clearly indicates to the engineer when the injector is operating, or if it happens to fly off, all without the necessity of watching the injector or its overflow from the cab window.

The Sellers system feeds the boiler with the same temperature of water when the locomotive is under load, standing, or drifting; its operation may be continuous, as the boiler feed is not interrupted by the above change of conditions. Owing to the small amount of live steam taken from the boiler and the unusual amount of heat in the feed water, the strain upon the boiler flues, the work required of the boiler, and the weight of coal consumed, are all reduced.

The injector is of simple construction, all nozzles may be removed and renewed without disconnecting from the locomotive.

Increased Boiler Pressure

During the last few years quite a number of railroads have utilized higher steam pressure on their locomotives. One railroad has over 500 locomotives which are using 250 pounds steam pressure, another road is now using the same steam pressure on a large number of locomotives, and still another road has now in service one locomotive which has a boiler carrying 350 pounds steam pressure, and this road is now contemplating the building of 25 more locomotives of the same design and carrying the same pressure. There is no question regarding the theory that increased steam pressure will give greater fuel economy. The amount of heat, and consequently fuel, required to raise the steam pressure from about 200 pounds (which is the usual pressure carried on modern locomotives) to 250 pounds, or 350 pounds, is comparatively small. The increase in power, however, by the use of this increase in pressure is very large. Therefore, a considerable increase in steam pressure is desirable as it will give considerably greater power with a very small increase in the amount of coal burned.

There are quite a number of details, however, which must be considered when a steam pressure higher than 200 pounds is used. One of the most important is the matter of boiler design. On the locomotives used by one road referred to above, and which carry 250 pounds steam pressure, the thickness of the plates in the boiler shell are $1^{1}/_{4}$ inches. Plates of such thickness are hard to shape into a locomotive type boiler and the weight of the boiler built of such plates is excessive. If higher pressure is used, much heavier plates would have to be used and it is a question as to whether it would be advisable to build a boiler of this type with higher pressures than about 300 pounds.

If the higher pressures are used, it will be necessary to resort to some different design of boiler, probably something in line with what was used on one road now having a locomotive carrying 350 lbs. steam pressure. This boiler should be considered as a semi-water tube boiler having a barrel at the front portion, but which barrel is very much smaller in diameter than the usual type locomotive boiler construction. We further believe that if higher

caused by distortion of the boiler plates, the stresses in the outer fibres of the bolt material are so high as to cause the breakage under the head.

It is believed it will be conceded that, under such adverse conditions, no bolt may be expected to render long service.

This condition actualy exists on a large number of boilers and we feel certain that none of you are undecided as to the results that may be expected from such conditions and that, after convincing yourselves of such conditions, that there will be no doubt in your minds as to the necessity for a positive and constant indication of the soundness of the bolts being used to stay such boilers.

The remedy obviously is to correct the error in construction. The best evidence of this fact is that boilers of like design and even larger in size have been in service for the past ten years and are still in service with no difficulty being experienced from bolt breakage.

Having shown in the foregoing the necessity for a better and more positive means of inspection of locomotive boiler flexible staybolts, an explanation will be given of the application of telltale flexible staybolts and a method for their inspection, which provides a maximum of safety and efficiency with a minimum of maintenance and inspection costs, which also eliminates a large loss of engine service consumed by the present stripping and cap removal method.

Under the old cap removal method, the condition of the bolt can only be determined or known definitely on the particular day or days that the cap is off and under unfavorable conditions such as previously described even then it may be doubtful. During the remainder of the two year interval between cap removal, little or no knowledge is available as to the condition of the bolts. Some inspectors claim to be able to detect broken flexible bolts under hammer test without applying pressure to the boiler. Having had thirty-five years' experience in boiler construction and maintenance, I would caution responsible officials against placing too much credence in such claims.

The telltale flexible bolt is identically the same as the Tate bolt described heretofore, with the addition of a telltale hole extending through the entire length of the body section and terminating within the head of the bolt. The walls of the telltale hole are copper plated to prevent rust or corrosion within the hole and to prevent them from becoming closed from this cause. They are applied exactly the same as the ordinary flexible bolt. If the method of riveting closes the end of the hole, it may be reopened easily and quickly, after which a porous fireproof closure is applied that will prevent the accumulation of foreign matter from the telltale hole, and that will permit leakage of steam or water in case of a break or fracture, which serves as a daily indicator of the condition of the bolt. This feature provides a daily or constant, positive indication by this method, as compared to questionable information once in two years by the old cap removal method.

In addition to depending on the leakage through the telltale hole, an inexpensive method of periodically checking up the condition of the telltale hole is provided as follows:

The fireproof porous closure is first removed, after which the specially constructed testing instrument is inserted. Upon reaching the extreme end of the telltale hole and making contact therewith, a light flashes in the handle of the tester indicating that the hole is open and therefore operative throughout its entire length.

The method of testing has been built upon the fact that a broken bolt having a telltale hole will show leakage of water or steam, providing the telltale hole is open and operative and that it extends to every part of the bolt.

The tester is so designed and constructed that it will positively indicate whether or not the telltale holes are open throughout their entire length. After inserting the tester in each telltale hole and securing light in the handle (which indicates that contact has been made with the extreme inner end of the hole) if the bolt is broken, or fractured into the hole, leakage will positively occur when water pressure is applied to the boiler.

Whentesting for broken staybolts first break through the porous closure with a sharp pointed pin or punch and a light hammer, then blow all of the remaining particles out of the telltale hole with the tool provided with the testing equipment for that purpose. Attach the ground connection in any convenient telltale hole, then insert the test rod into each telltale hole until contact with the end of each hole is secured. Such contact is indicated by the lighting of the bulb in the tester handle.

After contact has been secured in each and every bolt, apply water pressure to the boiler and every defective bolt will be indicated by leakage through the telltale hole. If no defective bolts are found, or after replacing any that are found, again close the telltale hole with the fireproof porous material and the engine is ready for service.

Cases may occur where breakage or fracture will be indicated by leakage and will not be observed or detected at the time they develop, as for instance in engines in pusher service at isolated points or bolts located behind brick arches, grate bars, etc., and in which the telltale hole will gradually become filled by accumulation of scale from the boiler water. Therefore, whenever the tester is inserted, it strikes an obstruction and fails to show a light in the handle. In such cases the tester should be removed and a special cleaning drill applied to remove the obstruction. After drilling, blow clean with the air tool, re-insert the tester, and if the hole has been thoroughly cleaned, contact will be secured and indicated by the lighting of the bulb as before described, and when water pressure is applied, leakage will occur.

Securing contact in the telltale hole does not indicate that the bolt is in good condition, but only that the tellale hole is open and operative throughout its entire length. It is the failure of the bolt to leak under pressure after contact has been obtained which indicates that it is not broken.

With this description of the method and use and comparison of results, we will now proceed to a comparison of costs. These comparisons are based on actual timed tests by both the old and new methods and the figures quoted are accurate and authentic.

The present method of inspection requires from three to four or more days, the principal part of the work being the removal and replacement of parts, rather than the actual time required by the inspector to examine the bolts.

By the new method herein described, it is not necessary to touch or remove anything on the outside of the boiler and the entire test on a modern locomotive boiler containing a full installation of flexible bolts can be completed within an eight-hour day—and at a labor cost of from $10 to $20, depending on the size of the installation.

The cost to strip, remove caps, inspect, and replace runs from $100 to $250, depending upon the size of the locomotive and upon the facilities at hand, and in addition, results in two, three, and sometimes more, days loss of engine service.

A number of railroads have been using the new method of inspection just described for some time, irrespective of the fact that the government had not yet approved